

GhostProofJob - OUTSTANDING INDEX (single reference sheet)

Purpose: the ONE place to see every open item by its ID, its status, and where the detail lives - plus a step-by-step manual-test checklist the founder runs. Check items off (- [] -> - [x]) as we complete them.

Current live build: v246 (CACHE_VERSION = gpj-v246; live == repo, verified 2026-08-27).

Companions: sprint-roadmap.md (per-item detail), BUILD_HISTORY.md (per-build log), guardrails.md (rules), feature-audit.md + launch-readiness.md (older P0/P1/P2 study).

Maintained by hand each change (the roadmap is auto-noted by bump_version.py; this sheet is curated).

0. THE RULES WE WORK UNDER (self-contained - never bypass)

- (!) **CL / CI RED = BLOCKING.** Any red CI/test status is a blocking failure, **never** a "flake."

A red run is **stop-and-fix at the root cause** (precedent: v156 rater race made deterministic, not retried).

- **INSERT-ONLY.** Add narrowly; never rebuild/restructure/redesign working code or layout.

• [UI-REVIEW] **gate.** ANY change to layout / a view / an overlay / z-index / visual behavior / a core flow -> STOP, propose the approach + mockup, get explicit approval **before writing code.**

• [STATE-COVERAGE] **before code.** Map the 4 quadrants - Guest - Authenticated - Interrupted-network - Empty-data - and add a Playwright test for any uncovered state.

- **Data-write changes are feature-class ALWAYS** (5-part approval + exact rollback), even fixing a bug.

• **Ask before:** editing the Cloudflare Worker - changing firestore.rules - consuming new quota - deleting/renaming any file.

• **The Matrix Gate (no skips), every change:** (1) node scripts/benchmark.mjs -> **BENCHMARK GREEN** (JS syntax - boot harness "RAN TO COMPLETION" - <div> delta 0 - mirror byte-identical - no dup DOM ids - on* audit - 3 version markers in sync - <=12 Vercel functions - case-exact requires - sitemap <=50k) -> (2) 8 backend suites + pool:check + rules emulator (Java/CI) -> (3) full Playwright chromium+mobile, all pages x light+dark x in/out x desktop/mobile x orientation, **0 failed / 0 flaky** -> (4) founder in-app self-test.

• **Live post-deploy check.** After every deploy, verify the LIVE url (ghostproofjob.com) is the new build (APP_VERSION/build-stamp == repo) - the local gate can't catch a stale deploy (it happened once).

• **Honesty over optimism.** No fabricated data (- when not real); honest "jump to apply" (auto-apply is impossible); no scraping; every AI action says when it fell back to templates AND when live AI returned.

- **Full drop-in files** for every changed file - never snippets. --workers=2 Playwright (never raise).

• **Deploy flow:** main auto-deploys to Vercel - **get explicit go-ahead before pushing.** Rollback prefers git revert <sha>. index.html re-uploads are **drag-and-drop only** (pasting truncates the ~1.5MB file).

1. THE ONLY HARD DEADLINE - Firebase credit clock

[] **E2 - D1 read-cost reduction - MUST land before 2026-09-19 (Firebase Blaze trial credit expiry).**

Screenshot 2026-08-27: **\$274.43 credit left - 23 days.** After it lapses, usage over the **free tier (50K reads + 20K writes/day)** bills the card. Biggest sink already cut (job_pools pool: deck reads ~3,800->~6). **Remaining:** session-cache Browse + company-view reads, cap query sizes, paginate; check reverse-match nightly + Ghosts read

volume. Founder budget cap ~ \$20 total. **Read-cost audit DONE 2026-08-27** -> docs/E2-read-cost-audit.md.

Ranked fixes (read-path only; no [UI-REVIEW]; each ships a state-coverage test):

- [] **E2-1 - Lower live-fallback caps + pool-freshness signal** - removes the surprise-bill tail risk (a stale pool makes every fetch a 3,000-8,000-doc live read). **Do first.**
 - [] **E2-2 - Company view reuses the in-memory pool** instead of loadCompanyJobs (<=800 reads/open) - biggest normal-path saver.
 - [] **E2-3 - Session-memoize ghost counters** (countJobReports/countGhostReports read <=200 docs each; the job counter fires on every card paint).
 - [] **E2-4 - Two-slot session cache + longer TTL** (single slot thrashes on region<->nationwide; 10-min TTL re-reads).
 - [] **E2-5 - Cache the per-fetch internal-jobs query** (limit(300) on every fetchJobs; minor until the employer side grows).
- [] **Founder step:** screenshot **Firestore -> Usage** (reads/day + trend) - tells us if we're already under 50K.
- [] **Founder safety net:** set a **Firestore budget alert/cap** (Billing -> Budgets & alerts) so nothing surprises you.
- [] **E2-P - Stale-job PRUNE is fragile / likely not firing** [BUILD][FEATURE-CLASS: data-delete -> needs 5-part approval].

The harvester HAS a prune (prune_stale_jobs, STALE_DAYS=8, deletes by ingestedAt), but it only runs on a "verify day" (cycle_day==7) AND **not** during STACKING mode - and recent stacking runs (pool build #61) plus the Usage screenshot (**Deletes = 0/7 days**) say it hasn't fired. Design risk: the ~205K-job pool may not be re-scraped within 8 days, so a live job could be wrongly pruned. **Proposed fix (needs founder sign-off + rollback):** run the prune every non-stacking harvest; set STALE_DAYS to safely exceed the real coverage-lap time; consider active:false marking before hard-delete. **Founder step:** open the latest Daily Job Harvest run -> paste the cycle_day=... verify=... pruned=... line (or authorize gh) to confirm. Detail: docs/E2-read-cost-audit.md context.

2. NEW - found in the 2026-08-27 live audit (not yet in the roadmap)

- [] (!) **N11 - P0 - Daily Job Harvest is FAILING (jobs going stale).** [BUILD][INFRA][data-write]

Run #76 (8/25) crashed: google.api_core.exceptions.InvalidArgument: 400 Invalid database id (default) at batch.commit(); today's run was **cancelled at the 60-min timeout**. Root structure confirmed: init_firestore is correct, but the **google dependency stack is UNPINNED** - firebase-admin==6.5.0 (2024) now auto-pulls google-cloud-firestore 2.29.0 + google-api-core 2.34.0, so the runner's deps drift daily. (The routing-header paren-encoding I first suspected is normal/old - a red herring.) ****Cannot confirm transient-vs-persistent from the sandbox (no backend creds; local Python is 3.14, runner is 3.12).** Plan: **(A)** manual re-run now** (free, definitive) -> tells us if it recurs; **(B)** **pin the google stack** in requirements.txt regardless (stops silent drift) - verified by the CI run; **(C)** **timeout hardening** (a hung commit-retry can't eat 60 min); **(D)** ****failure notification**** (see N12). Impact: no fresh jobs since ~8/24 -> pool content ages; the app may drift to the live-query read-path (explains reads at 32K).

Consequence for E2-P prune: even on 8/25's verify-day the prune never ran - the harvest crashed before reaching it.

[] **N12 - No failure alerting on daily actions.** [BUILD][INFRA] Nothing tells the founder (or Claude) when a cron workflow fails - it died quietly for days. **Fix:** add an if: failure() step to job_harvest.yml (+ the other crons) that emails via the existing **Resend** wiring (and/or opens a GitHub issue) with the failing step + error. Native, no inbox scraping. Optionally a daily pool-freshness health check (builtAt < 36h).

- [] **N1 - Wrong company logo = LinkedIn "in" glyph (systemic).** [UI-REVIEW] - HIGH-visibility, honesty.

On the **swipe card** AND the **Ghosts company list** (both themes), companies sourced from LinkedIn show the LinkedIn logo, because companyWebsite = a linkedin.com/company/... URL and _gpjLogoDomain (index.html ~14985) has **no job-board/aggregator host exclusion**. **Fix:** blacklist linkedin/indeed/glassdoor/ziprecruiter/monster/greenhouse/lever/ashbyhq/myworkdayjobs/icims/google/bing -> return '' -> falls to the

brand-map -> existing / **placeholder**. Single-point fix; add a state-coverage test.

[] **N2 - "98% Match" on every Browse card**. Live-confirmed the tracked over-generous score - Murray/Dow/Conroe/Houston Methodist all read 98% while the relevance chips read 12%/26%/8%. Same-field over-generosity was never tightened (v212/v216 fixed cross-level + messaging only). **Fix:** tighten computeMatch/scoreCore generosity so a weak in-field fit isn't in the 90s. (This IS the Sprint-B "Match %-honesty" item.)

[] **N3 - Ghost-page empty-state copy**. [UI][DECISION] Replace bare "-" with "a report is logged; a community % appears once there's enough signal"; relabel "hiring near your search" companies that have ****zero live roles**** when tapped. (Same as the roadmap "Ghost-page intuitiveness pass".)

[] **N4 - Mobile tap targets < 44px**. [UI-REVIEW] Sign In (28px), Support Us (28px), Match to Job (30px), Cover Letter (30px), filter chips (29px) are below the 44px iOS / 48dp Android minimum. Padding-only fix; add a mobile Playwright assertion (primary buttons >=44px tall).

[] **N5 - Cross-surface brand-purple mismatch**. App light - -cyber = #7C3AED but Resources pages (resources/index.html) use #7A3CA8. Unify (folds into F-WORDING/B3).

[] **N6 - "Resume Optimizer" label appears twice** in Résumé Studio (collapsed header + expanded block reuse the same title). Minor copy/polish.

[] **N7 - Repo cruft** - delete the stray empty ran file and \${OUT}v246_salary_tiles.png (leaked shell var in a filename) at repo root. (Trivial; "ask before deleting" - confirm.)

[] **N8 - Automate the live-URL post-deploy smoke** (= E3 below) - the local gate can't catch a stale deploy.

[] **N9 - Admin Diagnostics - readable Bug & Error report** [UI-REVIEW][BUILD] - founder-requested.

BUILT v247 (2026-08-27) - awaiting deploy + founder test T12. Benchmark GREEN - 8 backend suites PASS - N9 state-coverage 4/4 - smoke 12/12 - both-theme visual proof captured. Full Playwright running. Not yet pushed (needs go-ahead). Today the admin panel shows only the 24h client-error COUNT (adminLoadErrCount), and the actual data lives where the founder can't easily read it: client_errors docs {msg, src, line, ua, v, ts} (admin-read; auto-captured from window.onerror/unhandledrejection) are only viewable in the **Firestore console**, and user bugReports (their text + a 10-line consoleTrail + context) are delivered to **support@ghostproofjob.com** (Worker /contact always sends to support) and stored in Firestore bugReports - see N10 re: the reply-to. **Build:** an admin-only "Diagnostics" report under the existing admin panel that lists BOTH - (a) **client errors grouped/deduped by msg|src|line** with occurrence count, latest time, app version, browser; (b) **user bug reports** (description + console trail + timestamp) - each in plain, readable language, plus a **"Copy report" button** so the founder can paste it straight back here to resolve. **No rules change needed** (both collections are already admin-read). **[FREE-TIER]:** admin-only, manual-trigger, **capped** (e.g. last 100 errors + last 50 reports) -> a bounded handful of reads only when opened, never on the candidate hot path - consistent with the existing count-only design. **[STATE-COVERAGE]:** admin (renders) - non-admin (hidden/denied) - empty (honest "no errors/reports in range") - read-fail (message, never blank).

[] **N10 - Bug-report reply-to is a dead inbox**. [BUG] **BUILT v247 (folded into N9)**. The bug-report send (index.html:15826) sets the

Worker /contact payload email: 'bugs@ghostproofjob.com' - bugs@ **is not an active mailbox**. The report itself IS delivered (Worker always sends to support@; it's also written to Firestore bugReports), but email becomes the **reply-to**, so replying to a bug-report email bounces. **Fix:** set the reply-to to the **signed-in reporter's email when available, else** support@ghostproofjob.com - never the dead bugs@. One-line, non-visual, no Worker/rules change. Good to fold into the N9 build.

3. OPEN WORK BY SPRINT (Sprint A is [x] complete)

Sprint B - Trust the Intelligence (verify + harden the core value)

[] **B1 - Cover-letter / AI quality** (F-AI, F-COVERLETTER) - verify end-to-end now the Worker is redeployed;

fix any unfilled phrasing ("the this role position") / forced emphasis; confirm honest fallback labels + per-tier caps. Live-quality is a founder gate (Worker prompt is outside the repo).

- [] **B2 - Rater + ATS-preview** (F-RATER, F-ATSPREVIEW) - confirm the rater reads the WHOLE résumé, scores on

the stable 7-day corpus, two honest labelled scores; confirm the ATS preview shows the REAL parsed data (never audited - launch-readiness open Q3).

- [] **B2b - Match %-honesty** (= N2) - tighten same-field generosity; cross-field cap holds.

[] **B3 - Site-wide wording / pricing consistency sweep** (F-WORDING) - one honest story everywhere (app + static + Resources + checker); folds in N5 brand-purple unify.

[] **B-misc - Spell-check reachability** - `resumeSpellCheck()` runs only at import; "Improve My Whole Resume" doesn't spell-check and there's no button (feature-audit A3). Decide: wire it into Jett / add a button.

Sprint C - The Wow Pass (all [UI-REVIEW]; C1/C2/C3 shipped)

[] **C4 - Motion & delight (applicant only)** - swipe spring/touch-drag physics, count-up stats, streak flame, extend Apply/Hired celebrations. Employer side stays calm; respect prefers-reduced-motion; never block the core action.

- [] **C5 - Skeleton loaders + mascot empty/first-run states** - cheapest "feels fast + finished"; kill blank flashes.

[] **C6 - Signed-out home hero + honest social-proof bar** - first thing a new user sees; muted looping demo swipe; aggregate numbers ONLY when real (ties to F-GHOST data).

Sprint D - Signature Features (3 of 4 already built -> VERIFY; D2 is the one build)

[] **D2 - Full Inbox tab** [UI-REVIEW] [BUILD] - replace the interim per-message dismiss (v196) with a real inbox (anti-ghosting record + employer + candidate in one place). Founder: important, launch-window.

- [] **D1 - F-GHOST aggregated flag counts** - VERIFY the count aggregates with real volume + the "another hunter

reported this" popup surfaces everywhere.

- [] **D3 - 5 résumé templates** - VERIFY each exports cleanly (accent/headshot/spacing/address toggles intact).

[] **D4 - Broaden-location flow + "other regions" control** - VERIFY the same-state->statewide->other-cities ladder + B-SALARY-CYCLE (client-side salary filter) + B-SARATOGA hard-scope.

Sprint E - Growth & Cost

- [] **E2 - D1 read-cost reduction** - see §1 (the deadline).

[] **E3 - F-TEST hardening / signed-in CI + live-URL post-deploy smoke** (= N8) - authed Playwright in CI + a Playwright job against the production URL (home loads, deck fetches, `/smart-match` responds, `APP_VERSION==HEAD`).

- [X] **E1 - Resources cron GO-LIVE** - [x] LIVE (v243 go-live; every-other-day publisher). Monitor output.

Company real-data (cross-cutting)

[] **Company logo/website from the apply-URL domain** [BUILD] - derive the employer domain from a direct-employer "View Full Posting" URL (e.g. `jobs.geisinger.org->geisinger.org`) for the logo + a real Website button; exclude aggregator/ATS hosts (shares the N1 blocklist). Fixes Geisinger's missing logo.

- [] **Harvester long-tail logos** - `company_url->companyWebsite` (v240) populates as the pool self-heals (~8 days). Verify on a live harvest.
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4. CARRIED-OVER P1/P2 (from the v157 launch-readiness study - de-duplicated)

Resolved since v157 (verify live, then keep checked):

- [X] Node 20->24 (P0-1) - engines:"24" + all workflows on 24. [x]
- [X] Hide Ledger 60-cap (P0-2) - v186 (~5,000 keys, cloud-synced). [x]
- [X] Matching ignores education/certs (P1-4) - v187. [x]
- [X] Rater benchmarks wrong role (feat-audit A1) - v156 targets the wanted role. [x]
- [X] Public résumé checker (P2-3) - shipped /resume-checker.html (zero-read, static). [x]
- [X] Message hide-for-me (P1-6/D6) - v196 per-message dismiss. [x]

Still open:

[] **P1-2 - Reverse match returns nothing** - blocked on the FIREBASE_SERVICE_ACCOUNT secret / one run-log line (open since v145). **Founder action.**

[] **P1-3 / G7 - Digest emails x3** - approved, unbuilt (cheap now the pool is live).

[] **P1-7 / F7 - Hire data captured but not surfaced** - do NOT publicly claim hire numbers until an aggregate view exists.

[] **P2-5 - Offline queue** - an action taken offline may never reach the cloud (unknown/unbuilt).

[] **P2-6 - "My Data" audit view** - let users see what's stored (unbuilt).

[] **P2-7 - Backup branch auto-update on green** - the deep backup branch is far behind (intentional fallback); automation approved, unbuilt.

[] **P2-8 - Speed Insights re-add + guard** - was merged then silently lost in a full-file rewrite.

[] **P2-10 - Dead files** - manifest.json (unreferenced; note manifest.webmanifest is the live one) + frontend/SwipeCardQuery.js. **Ask before deleting.**

[] **SEO (P2-1/P2-2 / G1)** - 312 templated pages "discovered, not indexed" (crawl reality, PARKED, not a bug); duplicate company pages - generator doesn't fold name variants like the app's _coKey.

[] **Dead-code / orphan audit** - openCardCompanyProfile (CLAUDE.md §6 landmark, zero callers) + ~9 other unreferenced fns; confirm retired vs. lost.

5. FOUNDER MANUAL ACTIONS (external consoles - I can't do these; confirm status)

[X] Deploy latest **Cloudflare Worker** - [x] done (latest deployed 2026-08-26/27, founder-confirmed).

[X] Firestore **point-in-time recovery** - [x] enabled.

[] **Confirm** firestore.rules **is deployed** - D1 pool reads it live (tracker says LIVE); confirm the console shows current rules.

[] **DMARC** TXT record for Resend + confirm **SPF/DKIM** green (P2-9). _dmarc TXT: v=DMARC1; p=none; rua=mailto:support@ghostproofjob.com; fo=1.

[] Confirm Worker emails use support@ (not noreply@) - if the latest Worker deploy included it, check this off.

[] **Branch protection** on main (require verify) + stable (CI-only push, no force).

[] **Set a Firebase budget alert/cap** (the credit-clock safety net - §1).

[] Resolve the **Cloudflare vs Vercel DNS** confusion (guidance only - I won't change DNS): decide the domain

stays on **Vercel DNS** (avoids moving nameservers to Cloudflare / its paid tiers). The Worker runs on `*.workers.dev` and does NOT need the apex domain.

- [] `FIREBASE_SERVICE_ACCOUNT` **GitHub secret** - set it (unblocks reverse-match P1-2).
 - [] **Expand the employer test listing past 26 chars** (P1-5) - so it clears the listing-strength pin gate.
 - [] **Walk account creation end-to-end** (signup -> profile write -> first-run) - never exercised by anyone.
 - [] **Confirm the recruiter test-account login** (was timing out).
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6. PARKED DECISIONS (need a founder call)

- [] **Button *word* vs *icon* centering** (10px, site-wide) - A: keep icon, pin left - B: drop icon. On hold; remind after Sprint C.
 - [] **Ghost-% placement** - v245 hid the card-face tile (kept the drawer one). Reversible in 2 CSS lines if you prefer at-a-glance on the card face.
 - [] **N3 Ghost-page decisions** - (a) do flagged companies still surface in your hunt? (b) exact "hiring near your search" relabel/hide rule.
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7. TEST: MANUAL TEST CHECKLIST - step-by-step (founder runs these)

Run on a **real device**, **logged in** unless noted, and check **both dark + light** (profile menu -> theme toggle) and **desktop + mobile** (resize the window / open on a phone). Hard-refresh first so you're on the latest build.

T0 - Confirm you're on the current build (do this first, every deploy)

- [] Open `ghostproofjob.com`. In the browser console type `APP_VERSION` (or check "What's New") -> it should read **v246**

(or the version we just shipped). If it's older, you're on a **stale/cached deploy** - hard-refresh (Ctrl/Cmd-Shift-R) or re-upload `index.html`.

T1 - In-app Self-Test (the fastest health check - 19+ checks)

- [] Profile chip (top-right "Aaliyah") -> **Run Self-Test** -> wait for it to finish -> **every row green**.

Any red = tell me the row name; that's a real failure, not a flake.

T2 - Company logos (the N1 fix - after we ship it)

- [] **Swipe** a few cards + open **Ghosts** -> company logos are either the **real company logo** or the **/ placeholder** - **never the blue LinkedIn "in" logo** on a non-LinkedIn company. Check dark + light.

T3 - Match % honesty (the N2 fix - after we ship it)

- [] **Browse** your list -> the green **"Match"** badge is **not 98% on every card**; a weak/out-of-field role reads a believable lower number. Open a strong match and a weak one -> the numbers differ sensibly.

T4 - Ghost-page intuitiveness (the N3 fix - after we ship it)

- [] **Ghosts** -> companies with no community data show the honest *****a community % appears once there's enough**

signal*** line, not a bare "-" with no context. Tap a "hiring near your search" company -> it either shows real open roles or is honestly relabeled (no dead "none found" surprise).

T5 - Mobile tap targets (the N4 fix - after we ship it)

[] On a **phone**: Sign In, Support Us, **Match to Job**, **Cover Letter**, and the filter chips are **easy to tap** (comfortably thumb-sized, ~44px tall) - none feel like a thin sliver.

T6 - Résumé tailoring / AI quality (B1 - needs the Worker deployed, which it is)

[] **Résumé** -> open a real job -> **Match to Job** -> the reframe touches **only 2-3 bullets**, reads natural (NOT "marketing" stuffed onto every line), and **never invents** a metric/employer/title. **Send me the PDF**.

[] **Cover Letter** on the same job -> no unfilled phrasing ("the this role position"); it names real strengths; if live AI wasn't reachable it **says so** (honest fallback banner).

T7 - Rater + ATS preview (B2)

[] **Résumé** -> **Rate My Résumé** -> two honest labelled scores (writing quality + role fit); the score doesn't swing wildly between visits. **ATS preview** -> what it shows matches your real résumé content (name, titles, skills).

T8 - 5 template exports (D3)

[] **Export Template Studio** -> try each of the **5 templates** (Classic ATS / Modern Split / Minimal / Corporate Grid / Creative Accent) -> each downloads a clean PDF; toggling accent / headshot / spacing / address doesn't break the layout. Check one in dark and one in light.

T9 - Broaden-location ladder (D4)

[] **Browse** -> the "Show other parts of [State] -> / Show all regions ->" pill widens **only when you tap it** (never on its own). The "Only jobs with posted salary" toggle **filters in place** (doesn't re-pull random regions).

T10 - Both-theme + device sweep (every build)

[] Walk **Swipe** - **Browse** - **Résumé** - **Ghosts** - **Account** in **dark**, then **light** -> colors match the brand (Midnight Plum / Mint / Cyber Purple), text is readable, nothing off-center or clipped. Repeat on **mobile**.

T11 - Deploy proof (after each deploy)

[] Re-run **T0** on the live site + re-run **T1 Self-Test** -> confirms the deploy is live and healthy, not stale.

T12 - Admin Diagnostics report (the N9 build - after we ship it)

[] On your admin account: **Settings** -> **admin panel** -> **Diagnostics** -> you see a **readable list** of recent client errors (grouped, with counts + which browser + which build) AND user bug reports (their words + console trail) - not just the count. Tap **Copy report** -> paste it here so I can resolve the specific issues. _Until this ships, view client errors in the Firebase console (client_errors) and bug reports in your **support@** inbox or Firestore bugReports._

8. How we work each item (the loop)

[UI-REVIEW] mockup + approval (if visual) -> [STATE-COVERAGE] matrix + test -> INSERT-ONLY build (full files) -> **full Matrix Gate, no skips** -> visual proof (before/after, both themes) -> you approve -> I push (with your go-ahead) -> **live-URL post-deploy check** -> update this index + sprint-roadmap.md + BUILD_HISTORY.md.

Last updated: 2026-08-27 - against live build v246.